Handout 10: The rules refresher

REMUS

Everyone knows how the pieces move. These are the rules that trip people up, and the question Remus asks for each one. The "Day" column says when the code that needs the rule is written.

Rule	What has to be true	What Remus checks	Day
Check	The side to move's king stands on a square the other side attacks	`in_check()` asks whether any enemy piece attacks the king's square	4
A legal move	After the move, the mover's own king is not attacked. This forbids moving a pinned piece off its line and walking the king into check	`legal_moves()` makes each move, asks `in_check()`, and undoes it	4
Castling, short or long	The king and that rook have not moved yet (the right is still in the `castling` field), the squares between them are empty, the king is not in check, and the king passes over and lands on no attacked square	All four conditions, then the king moves two squares and the rook jumps over it	5
En passant	A pawn has just moved two squares, and an enemy pawn stands beside it. Only on the very next move	The `en_passant` field names the square behind the pawn; the capture lands there and removes the pawn beside it	5
Promotion	A pawn reaches the last rank. It must become a queen, rook, bishop, or knight: never stay a pawn, never become a king	Four moves instead of one, one per piece	5
Fifty-move rule	Fifty moves by each side, 100 plies, with no capture and no pawn move	The `halfmove_clock` counts plies since the last capture or pawn move and resets to 0 on either; a draw at 100	7
Threefold repetition	The same position, with the same side to move, the same castling rights, and the same en passant square, appears three times	The position history holds one key per position; three equal keys is a draw	7
Stalemate	No legal move, and the king is not in check	Zero legal moves and `in_check()` is False: a draw, not a loss	7
Checkmate	No legal move, and the king is in check	Zero legal moves and `in_check()` is True: the side to move has lost	7
Insufficient material	Neither side can ever give checkmate: king against king, king and one minor piece against king, or king and bishop against king and bishop on the same color	Count the pieces left	7